Linear Free Energy Relationships. Part 1. Plots of Data

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*, Diagrams were edited in *Affinity Publisher*

The handout features several tables of data. I hope that you plotted them yourself and came to your own conclusions. Alas, we cannot discuss everything in a 50 minute class so here are my own plots and comments. You can use the Jupyter notebooks accessible via the course web book to make your own versions of these plots.

Table 4: Benzyl Chloride with Water and with Hydroxide Ion

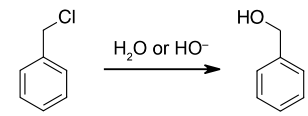

Figure 1: Hydrolysis of benzyl chloride

In table 4 of the class handout we present data for the rate of benzyl chloride reacting with water (in acidic conditions where hydroxide ion concentrations are near zero) and in various concentrations of aqueous sodium hydroxide solution.[1] The data from table 4 is plotted below. I chose to present the results for the highest concentration of hydroxide and for the acidic reaction. You can plot other data sets from table 4 by using the *Python* notebook available via Colab.

[1] Hydrolysis of benzylchloride in 50% acetone/water in the presence of acid and base. Data from "Relations Entre la Constitution du Substrat et sa Sensibilité à L'ion oh dans L'hydrolyse." S.C.J. Olivier, A.P. Weber, *Recl. Trav. Chim. Pays-Bas*, **1934**, *53*, 869–890. https://doi.org/10.1002/recl.19340531002

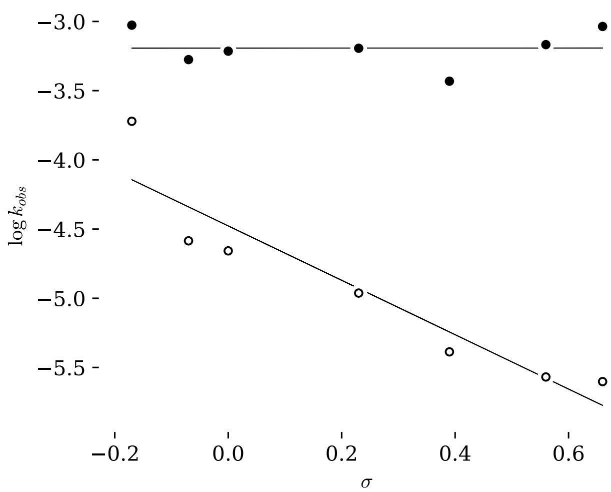

Figure 2: Hammett plots for acidic (0.27 M H_2SO_4, closed circles) and basic (0.26 M NaOH, open circles) hydrolysis of benzyl chloride

0.27 M H_2SO_4, $\rho = -0.00 \pm 0.19$, $R^2 = 0$

0.26 M NaOH, $\rho = -1.84 \pm 0.32$, $R^2 = 0.94$

Both reaction conditions result in S_N2 mechanisms and we expect a transition state structure that reflects a concerted reaction.

Table 6: Solvolysis of Diphenylmethylene Chloride

In table 6 of the class handout we presented data for the rate of diphenylmethylene chloride reacting with methanol, ethanol and isopropanol at 25 °C.[2] I chose to plot the data for solvolysis in ethanol. You can plot other data sets from table 6 by using the *Python* notebook available via Colab.

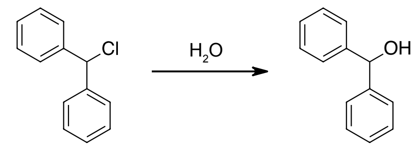

Figure 3: Solvolysis of Diphenylmethylene Chloride

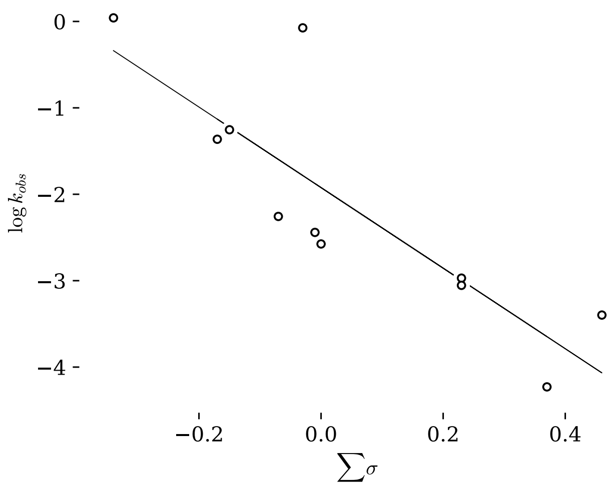

Figure 4: Hammett plot for solvolysis of diphenylmethylene chloride. Correlation is poor but we still see the trend. Perhaps we should use different σ parameters?

$\rho = -4.66 \pm 0.97$, $R^2 = 0.72$

[2] S. Altscher, R. Baltzly, S.W. Blackman, *J. Am. Chem. Soc.*, **1952**, *74*, 3649–3652. https://doi.org/10.1021/ja01134a054. J.F. Norris, C. Banta *J. Am. Chem. Soc.*, **1928**, *50*, 1804–1808. https://doi.org/10.1021/ja01393a049. J.F. Norris, J.T. Blake, *J. Am. Chem. Soc.*, **1928**, *50*, 1808–1812. https://doi.org/10.1021/ja01393a050.

The negative value for ρ indicates large positive change in charge in the transition state and the large magnitude of the value is consistent with a direct connection to the benzene ring. This indicates a carbocationic intermediate and an S_N1 reaction mechanism.

The poor correlation with Hammett σ values is the result of the carbocationic intermediate. the empty p-orbital is directly connected the the π-system of the benzene ring. *p*-Substituents that have lone pairs can donate electrons directly into the cationic centre and share the charge. this results is a much larger stabilization effect. We need a new set of σ values that reflect this property. In the next meeting we will plot this data agin using the Brown-Okamoto σ^+ values.

Table 7: Solvolysis of Trityl Chloride

In table 7 of the class handout we presented data for the rate of substituted trityl chlorides reacting with water in acetonitrile 25 °C.[3]

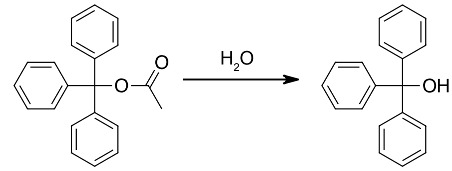

Figure 5: Solvolysis of Trityl Chloride

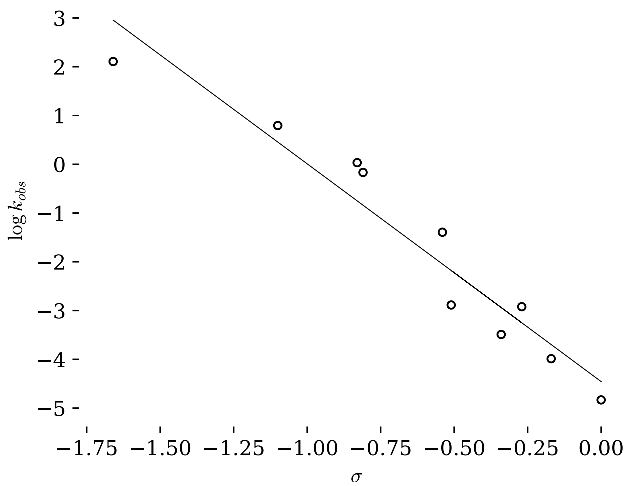

Figure 6: Hammett plot for solvolysis of trityl chloride. Correlation is good with Hammett σ parameters but, as we will see in the next meeting, we should have used Brown–Okamoto σ^+ values

$\rho = -4.47 \pm 0.44$, $R^2 = 0.93$

[3] "Stabilities of Trityl-Protected Substrates: The Wide Mechanistic Spectrum of Trityl Ester Hydrolyses." M. Horn, H. Mayr, *Chem. Eur. J.*, 2010, *16*, 7469–7477. https://doi.org/10.1002/chem.200902669.

I chose to plot the data for solvolysis in 90% acetonitrile. You can plot other data sets from table 7 by using the *Python* notebook available via Colab.

Analysis & Comments

Hammett plots can provide information about the transition state in a reaction. Let us consider the three cases above.

S_N1 Solvolysis from Data Tables 6 & 7

The plots for the solvolysis of diphenylmethylene chloride and triphenylmethylene chloride (trityl chloride) show that the mechanism involves the formation of a carbocation that is directly attached to the benzene ring. The reaction constants were negative and large in magnitude. In these cases it is more appropriate to use the Brown-Okamoto σ^+ parameters that include the effects of direct resonance stabilization of the cation by some substituents and we will discuss these two data sets in more detail in the following class meeting.

S_N2 Solvolysis Data from Table 4

In S_N2 substitution the transition state is not a carbocation. As electrons depart with the leaving group they are entering with the nucleophile and the electron density at the central carbon remains somewhat constant.

There is no carbocation to accept electrons from a resonance donor and so Hammett σ values would be expected to work well here.

From the plot for Table 4 we see that the Hammett reaction constant, ρ, for S_N2 is near zero. No significant charge difference occurs as we move from the starting material to the transition state. This is a nearly perfectly concerted reaction where approach of the nucleophile is synchronous with departure of the leaving group. Below is a More O'Ferrall–Jencks plot that describes this reaction.

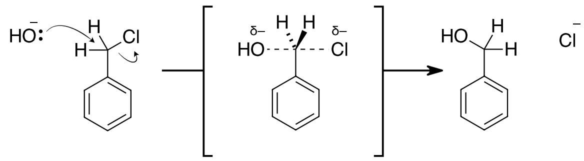

Figure 7: Mechanism of a concerted S_N2 reaction

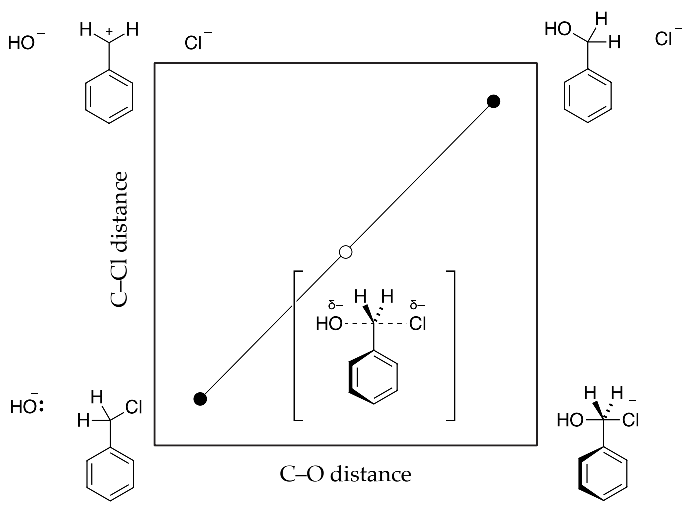

Figure 8: More O'Ferrall–Jencks plot describing S_N2 substitution with hydroxide ion and benzyl chloride.

In the case of water as the nucleophile we observe that $\rho = -1.8$. This indicates that the transition state is more positively charged at the reaction centre compared to the starting material. Obviously, since the intermediate is a positively charged protonated alcohol, we would expect the transition state to have some properties of the intermediate.

However, as the water nucleophile gains positive charge, the chloride leaving group gains negative charge. This would balance the charge in the transition state if the system was perfectly synchronous and we would see a small value for ρ. If we consider how the energy surface of the More O'Ferrall–Jencks plot changes as we change a strong hydroxide nucleophile for a weak water nucleophile, we see that the transition state is no longer positioned to describe a symmetrical structure. The bond to the leaving group is now longer in the t.s. This reduces the electron density at the centre carbon and creates positive charge on the atom directly connected to the benzene ring.

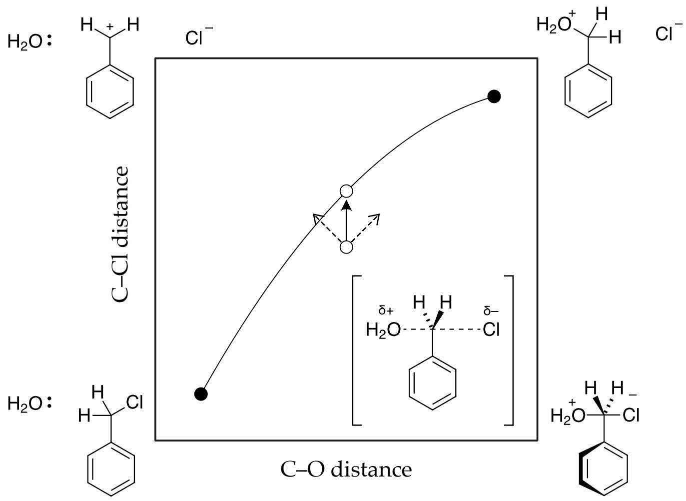

Figure 9: More O'Ferrall–Jencks plot describing S_N2 substitution with water and benzyl chloride.

So the Hammett plot confirms the hypothesis generated by the More O'Ferrall–Jencks plot. The t.s. is not symmetrical

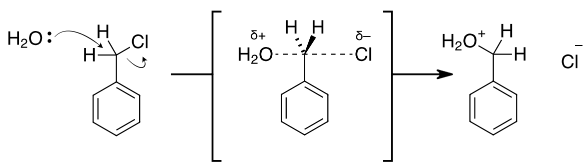

Figure 10: Mechanism of a concerted S_N2 reaction with an assymetrical t.s.

One could then predict that using a stronger nucleophile than hydroxide would move the position of the t.s. in the opposite direction. In this case the bond to the leaving group would be short in the t.s. compared to the hydroxide case. If the leaving group is departing later compared to the approach of the nucleophile then we will be squeezing electrons together at the reaction centre.

One experimental result that I found was for iodide anion acting as a nucleophile. Iodide is a stronger nucleophile than hydroxide ($I^- > Br^- > Cl^-$)[4] and so we expect to see evidence of a t.s. with increased electron density at the reaction centre. The Hammett plot shows that $\rho = 0.79$.[5]

[4] Recall that larger halogen ions are more polarizable and are better nucleophiles *and* better leaving groups.

[5] You can find the data in "Velocities of reaction of substituted benzyl chlorides in two reactions of opposed polar types." G. M. Bennett, B. Jones, *J. Chem. Soc.*, **1935**, 1815–1819. https://doi.org/10.1039/JR9350001815. Plot the data for yourself and confirm the stated value for ρ.

Conclusions

We see that Hammett plots reveal information about changes in charges in transition states. We can use these tools of test hypotheses about the t.s. for a given reaction.